A Partially Derivative-Free Proximal Method for Composite Multiobjective Optimization in the Hölder Setting

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Joint work with P.B. Assunção and D.R. Souza

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Derivative-free methods

In many practical optimization problems, derivatives are not available or are too expensive to compute.

These problems can be solved using derivative-free methods.

Recently, in the scalar case:

→ G. N. Grapiglia(2024)[1], proposed a derivative-free quadratic regularization method.

→ V. S. Amaral(2025)[2], proposed a partially derivative-free method for composite optimization.

Our work is motivated by these papers.

[1] Grapiglia, G.N.: Worst-case evaluation complexity of a derivative-free quadratic regularization method. Optimization Letters 18(1), 195–213 (2024)

[2] V. S. Amaral. (2025). A partially derivative-free cyclic block coordinate descent method for nonseparable composite optimization. Mathematical Modelling and Analysis, 30(3), 535–552

Composite multiobjective optimization

In our work[3], we study the following optimization problem:

$$\min_{x \in \mathbb{R}^n} F(x), \tag{1}$$

where $F : \mathbb{R}^n \to \mathbb{R}^m$ has its components given by

$$F_j = f_j + h_j, \quad j \in \mathcal{N} = \{1, \ldots, m\}.$$

We assume that $f_j : \mathbb{R}^n \to \mathbb{R}$ is a differentiable function and $h_j : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is a convex function.

An important instance of Problem (1) is obtained when h_j is the indicator function of a convex set $C \subset \mathbb{R}^n$. In this case, Problem (1) becomes the constrained multiobjective problem:

$$\min_{x \in C} \big(f_1(x), \cdots, f_m(x)\big).$$

[3] V. S. Amaral, P. B. Assunção, D. R. Souza. A Partially Derivative-Free Proximal Method for Composite Multiobjective Optimization in the Hölder Setting.

Composite multiobjective problems

Recently, composite multiobjective problems have been addressed in several works, for example:

→ Proximal gradient: proposed by Fukuda et al.[4].

→ A proximal gradient method with an explicit line search: proposed by Bello-Cruz et al.[5]

→ Generalized conditional gradient: proposed by Assuncão et al.[6]

[4] Tanabe, H., Fukuda, E.H., Yamashita, N.: Proximal gradient methods for multiobjective optimization and their applications. Comp. Optim. Appl.

[5] Bello-Cruz, Y., Melo, J.G., Prudente, L.F., Serra, R.V.G.: A proximal gradient method with an explicit line search for multiobjective optimization. Comp. Optim. Appl.

[6] Assunção, P.B., Ferreira, O.P., Prudente, L.F.: A generalized conditional gradient method for multiobjective composite optimization problems. Optimization

Composite multiobjective optimization

Definition of Pareto optimal: We say that x^* is Pareto optimal point for Problem (1) if there is no x such that

$$F(x) \leqslant F(x^*) \quad \text{and} \quad F(x) \neq F(x^*), \tag{2}$$

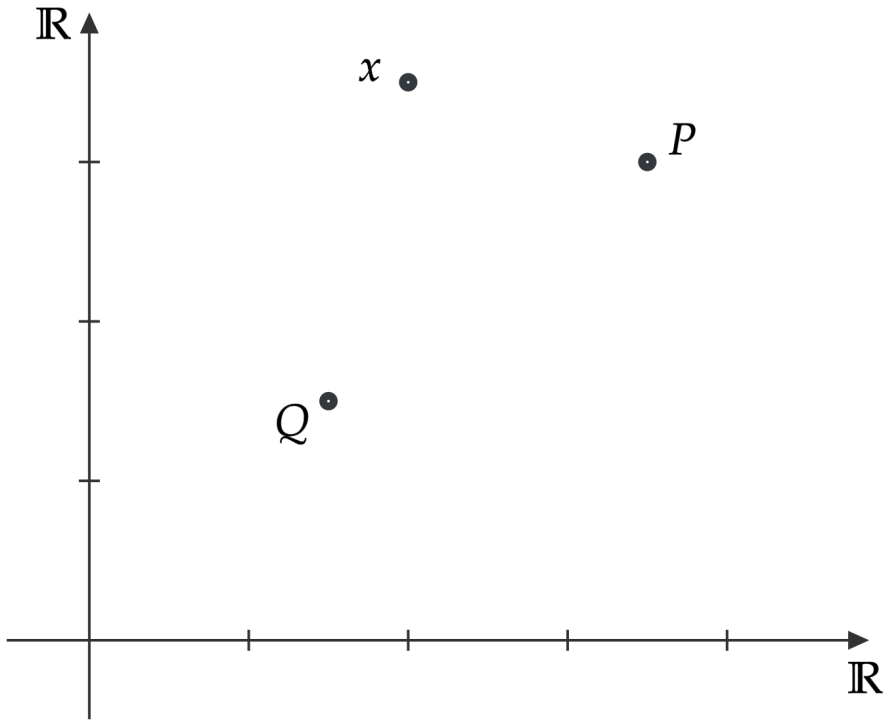
where the inequality $\leqslant$ between vectors is understood componentwise.

The condition (2) says that we cannot improve one objective without worsening the other.

Illustrative example

Example: $\min_{x \in \mathbb{R}^n} \left(F_1(x) = \frac{1}{2}\|x - P\|^2,\ F_2(x) = \frac{1}{2}\|x - Q\|^2 \right), \quad P, Q \in \mathbb{R}^n$

$\rightarrow$ Problem to minimize the distances to both P and Q.



Solutions on the line $P - Q$

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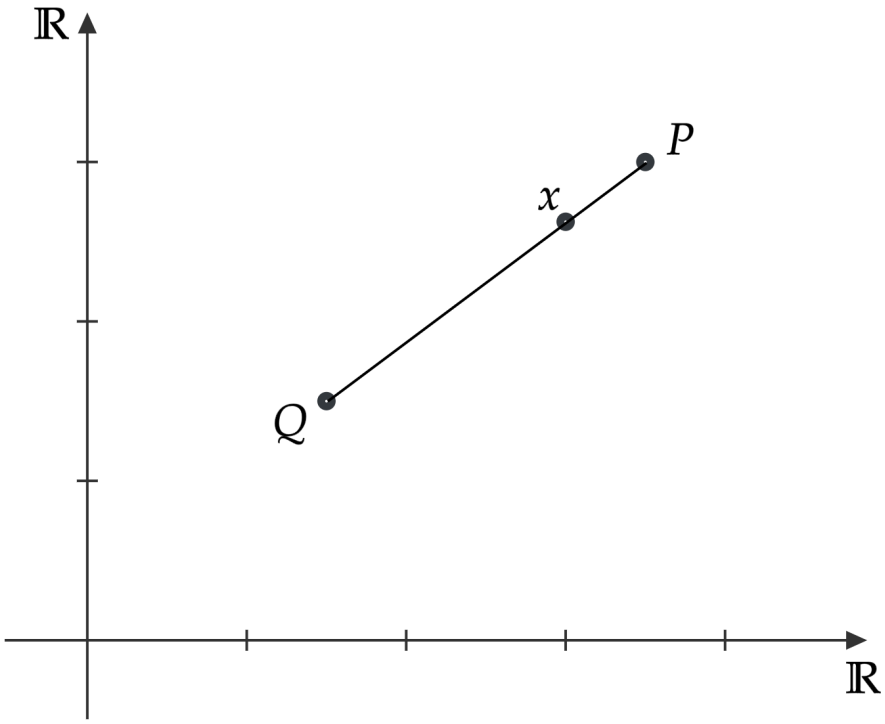

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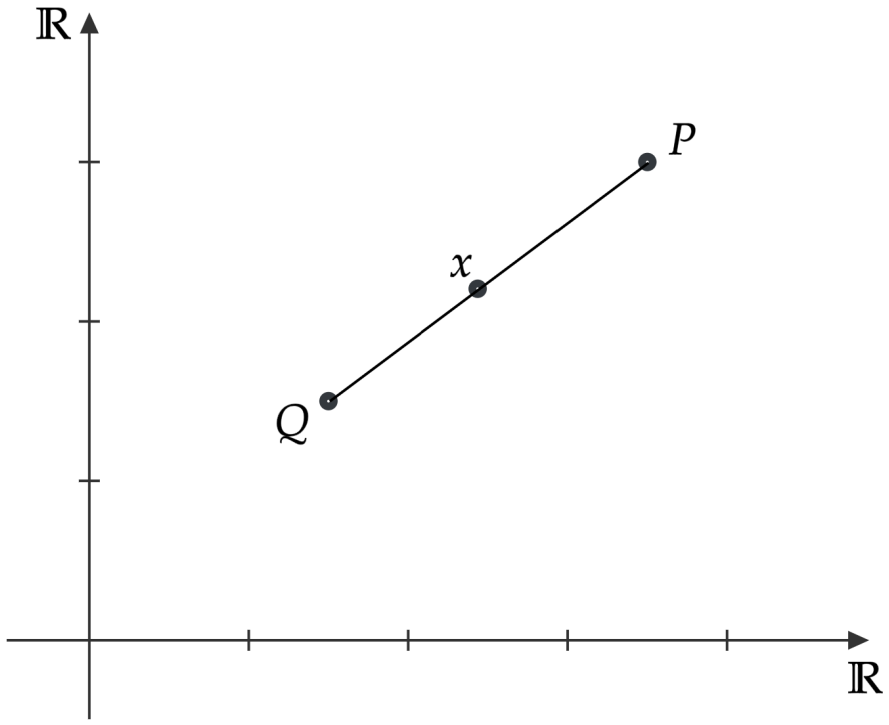
$\rightarrow$ Problem to minimize the distances to both P and Q.

- The solutions belong to the line segment that connects P and Q.
- On this line segment, it is not possible to get closer to Q without moving away from P..
- These points on the line segment are Pareto.



Solutions on the line $P - Q$

Partially Derivative-Free Proximal Method - PDFPM

We propose a Partially Derivative-Free Proximal Method - PDFPM to solve Problem (1).

Given x^k, the method PDFPM finds a new point x^{k+1} such that

$$F(x^{k+1}) < F(x^k).$$

Svaiter et al.(2000)[7] appear to have been pioneers in this approach for multiobjective optimization.

In each interaction, the method PDFPM minimize the following model:

$$\max_{j \in \mathcal{J}} \left[\langle g_{f_j}(x^k, \lambda_k) + \frac{1}{2} B_j^k (x - x^k),\, x - x^k \rangle + h_j(x) - h_j(x^k) \right] + \frac{\sigma_k}{2} \|x - x^k\|^2,$$

where $g_{f_j} : \mathbb{R}^n \times [0,1] \to \mathbb{R}^n$ is an approximation of ∇f_j and $B_j^k \in \mathbb{R}^{n \times n}$ symmetric positive semidefinite(SPSD).

[7] J. Fliege and B. F. Svaiter, *Steepest descent methods for multicriteria optimization*, Math. Methods Oper. Res. 51(3), 479–494 (2000).

Partially Derivative-Free Proximal Method - PDFPM

The PDFPM method

Algorithm: Let $x^0 \in \mathbb{R}^n$, $\alpha, \epsilon \in (0,1)$, $\sigma_0 \geqslant 1$ and a SPSD matrix $B_j^0 \in \mathbb{R}^{n\times n}$.

Step 1. For $\lambda_k \in \left(0, \frac{\epsilon}{\sigma_k\sqrt{n}}\right]$, compute $g_{f_j}(x^k, \lambda_k)$ for all $j \in \mathcal{N}$.

Step 2. Find $\bar{x}^k \in \mathbb{R}^n$ solution of the following subproblem:

$$\min_{x\in\mathbb{R}^n} \max_{j\in\mathcal{J}} \left[\langle g_{f_j}(x^k,\lambda_k)+\tfrac{1}{2}B_j^k(x-x^k),\, x-x^k\rangle+h_j(x)-h_j(x^k)\right]+\tfrac{\sigma_k}{2}\|x-x^k\|^2.$$

Step 3. If $\sigma_k\|\bar{x}^k - x^k\| < \epsilon$, Stop. Otherwise, go to ***Step 4***.

Step 4. If

$$F_j(\bar{x}^k) \leqslant F_j(x^k) - \frac{\alpha\epsilon^2}{2\sigma_k}, \qquad \forall j \in \mathcal{N}, \tag{3}$$

define $x^{k+1} := \bar{x}^k$, $\sigma_{k+1} := \sigma_k$, for each j choose a SPSD $B_j^{k+1} \in \mathbb{R}^{n\times n}$, $k \leftarrow k+1$, and go to ***Step 1***. Otherwise, $\sigma_k \leftarrow 2\sigma_k$ and go to ***Step 1***.

Well-defined

To ensure that the PDFPM is well defined, we assume the following assumptions.

Assumption 1. There exists a positive real number $\overline{B}$ such that

$$\|B_j^k\| \leqslant \overline{B}, \quad \forall\, k \in \mathbb{N} \text{ and } j \in \mathcal{N}.$$

Assumption 2. For each $j \in \mathcal{N}$, there exist positive real numbers L_j, M_j and $\beta_j \in (0, 1]$ so that the following conditions are satisfied.

$$\rightarrow\ f_j(y) \leqslant f_j(x) + \langle \nabla f_j(x), y - x\rangle + \frac{L_j}{2}\|y - x\|^2 + \frac{M_j}{\beta_j + 1}\|y - x\|^{\beta_j + 1}.$$

$$\rightarrow\ \|\nabla f_j(x) - g_{f_j}(x, \lambda)\| \leqslant \lambda \frac{\sqrt{n} L_j}{2} + \sqrt{n}\frac{M_j}{\beta_j + 1}\lambda^{\beta_j}.$$

Assumption 2 holds when some functions f_j have Lipschitz gradients, others have Hölder gradients, and g_{f_j} is obtained using finite differences.

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Well-defined

The following result shows that **Step 4** of the PDFPM is well defined.

Theorem: Let $\bar{x}^k$ the point obtained in **Step 2** of the PDFPM. Suppose that the Assumptions 1 and 2 hold.
If

$$\sigma_k \|\bar{x}^k - x^k\| \geqslant \epsilon \tag{4}$$

and

$$\sigma_k \geqslant \max_{j \in \mathcal{J}} \left[\frac{5L_j + \overline{B}}{1-\alpha} + \frac{2n^{\frac{1-\beta_j}{2}} M_j + 2M_j}{(\beta_j + 1)(1-\alpha)} \epsilon^{\beta_j - 1} \right]^{\frac{1}{\beta_j}}, \tag{5}$$

then

$$F_j(\bar{x}^k) \leqslant F_j(x^k) - \frac{\alpha \epsilon^2}{2\sigma_k}, \qquad \forall j \in \mathcal{N}. \tag{6}$$

In other words, there exists a σ_k that ensures a reduction in F in **Step 4**.

Thus, we ensure that the PDFPM method is well defined.

Complexity

Given a target $\bar{F} < F(x^0)$, we prove that the maximum number of iterations for which

$$F(x^{k+1}) > \bar{F} \quad \text{and} \quad \sigma_k \|x^{k+1} - x^k\| \geqslant \epsilon, \tag{7}$$

is given by

$$\max_{j \in \mathcal{J}} \left\{ F_j(x^0) - \bar{F}_j \right\} \frac{2c}{\alpha} \epsilon^{-\frac{\beta+1}{\beta}}, \tag{8}$$

where $c = 2 \max_{j \in \mathcal{J}} \left[\frac{2L_j + \bar{B}}{1-\alpha} + \frac{2n^{\frac{1-\beta_j}{2}} M_j + 2M_j}{(\beta_j + 1)(1-\alpha)} \right]^{\frac{1}{\beta_j}}$ and $\beta = \min\{\beta_j : j \in \mathcal{J}\}$.

It´s important to note that when β approaches zero, the number of iterations may increase.

In other words, the complexity worsens when β approaches zero.

Numerical Results

→ The numerical experiments were done in `Julia`;

→ The source code is available in the following QR code:

Figure: `https://github.com/Souza-DR/AAS2025-PDFreeMO`

Test Problems

→ We consider **composite optimization problems** of the form

$$\min_{x \in \mathbb{R}^n} \big(f_1(x) + h_1(x),\ \ldots,\ f_m(x) + h_m(x) \big).$$

Where:

- $f_j : \mathbb{R}^n \to \mathbb{R}$ is a differentiable function.
- h_j is a convex function defined by $h_j(x) := \max_{z \in \mathcal{Z}_j} \langle x, z \rangle$.
- The uncertainty set $\mathcal{Z}_j$ is a polyhedral:

  $\mathcal{Z}_j = \{z \in \mathbb{R}^n \mid A_j z \leqslant \delta e\}$, where $e = (1, \ldots, 1)^\top$, $\delta \geqslant 0$, and

  $A_j := \begin{bmatrix} \tilde{A}_j \\ -\tilde{A}_j \end{bmatrix}, \quad \tilde{A}_j \in \mathbb{R}^{n \times n}$ is nonsingular.

Remark: Here, h represents a particular case of a composite problem, known as robust optimization.

Comparison with Existing Methods

To evaluate the performance of the proposed algorithm, we compare it with existing methods for **multiobjective optimization**.

- → **PDFPM**: the proposed **Partially Derivative-Free Proximal Method**.
- → **ProxGrad**: a **proximal gradient method** proposed by Fukuda et al. (2019).
- → **CondG**: a **conditional gradient method** introduced by Assunção et al. (2023).

These methods are designed for composite multiobjective problems, so they are a natural comparison.

Numerical Tests

To illustrate the applicability of the proposed algorithm, we present the following two novel test problems.

→ **AAS1:** $\min_{x \in \mathbb{R}^2} \big(f_1(x), f_2(x)\big)$ where f_1 and f_2 are given by

$$f_1(x) = \tfrac{1}{2}\|Ax - b\|_2^2, \;\; f_2(x) = \tfrac{\mu}{p}\|Dx\|_p^p, \text{ with } \mu = 0.9 \text{ and } p = 1.003.$$

In this case, f_1 *has a* ***Lipschitz gradient****, while* f_2 *has a* ***Hölder gradient****.*

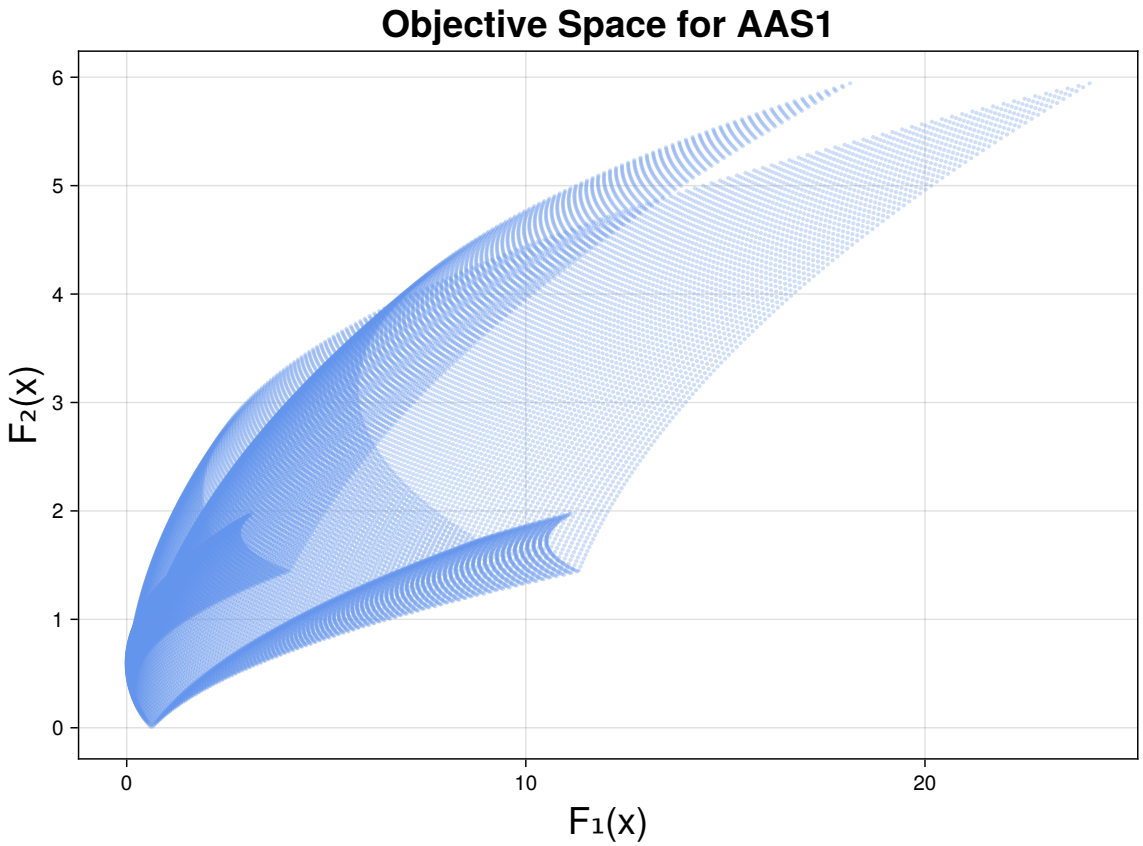
→ **AAS2** $\min_{x \in \mathbb{R}^2} \big(f_1(x), f_2(x)\big)$ where f_1 and f_2 are given by

$$f_1(x) = \tfrac{\mu_1}{p_1}\|D_1(x - c_1)\|_{p_1}^{p_1}, \; f_2(x) = \tfrac{\mu_2}{p_2}\|D_2(x - c_2)\|_{p_2}^{p_2}, \text{ with } \mu_1 = 1.2,$$

$\mu_2 = 0.8$, $p_1 = 1.003$ and $p_2 = 1.002$.

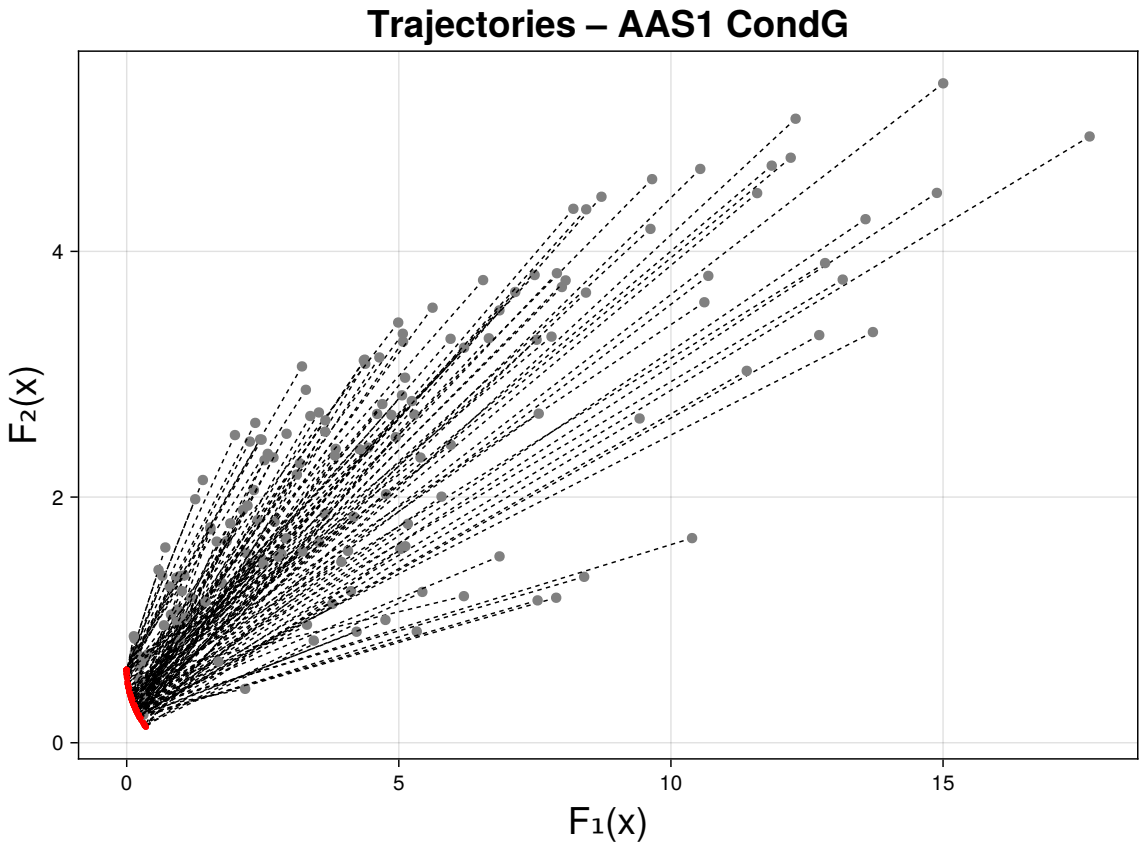
In this case, both f_1 *and* f_2 *have* ***Hölder gradients****.*

Pareto Fronts: AAS1



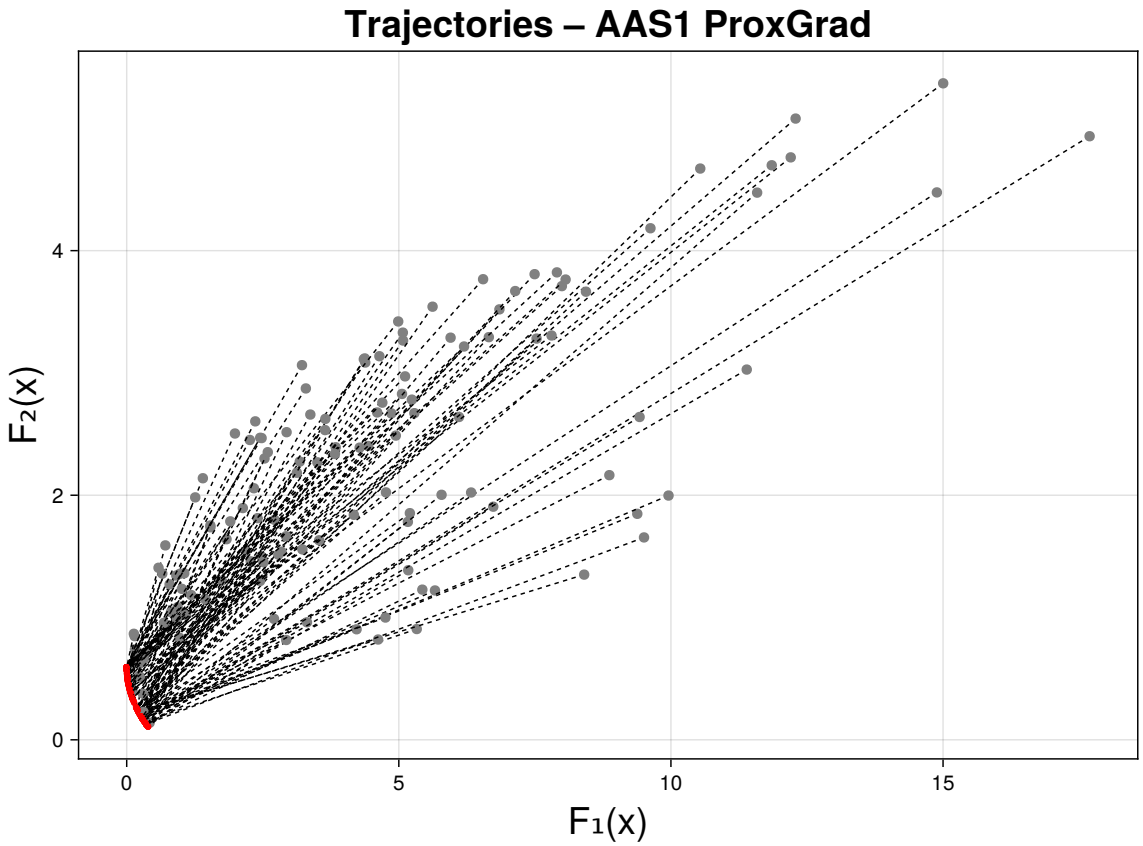
→ **Objective space:** global view of the biobjective problem.

Pareto Fronts: AAS1



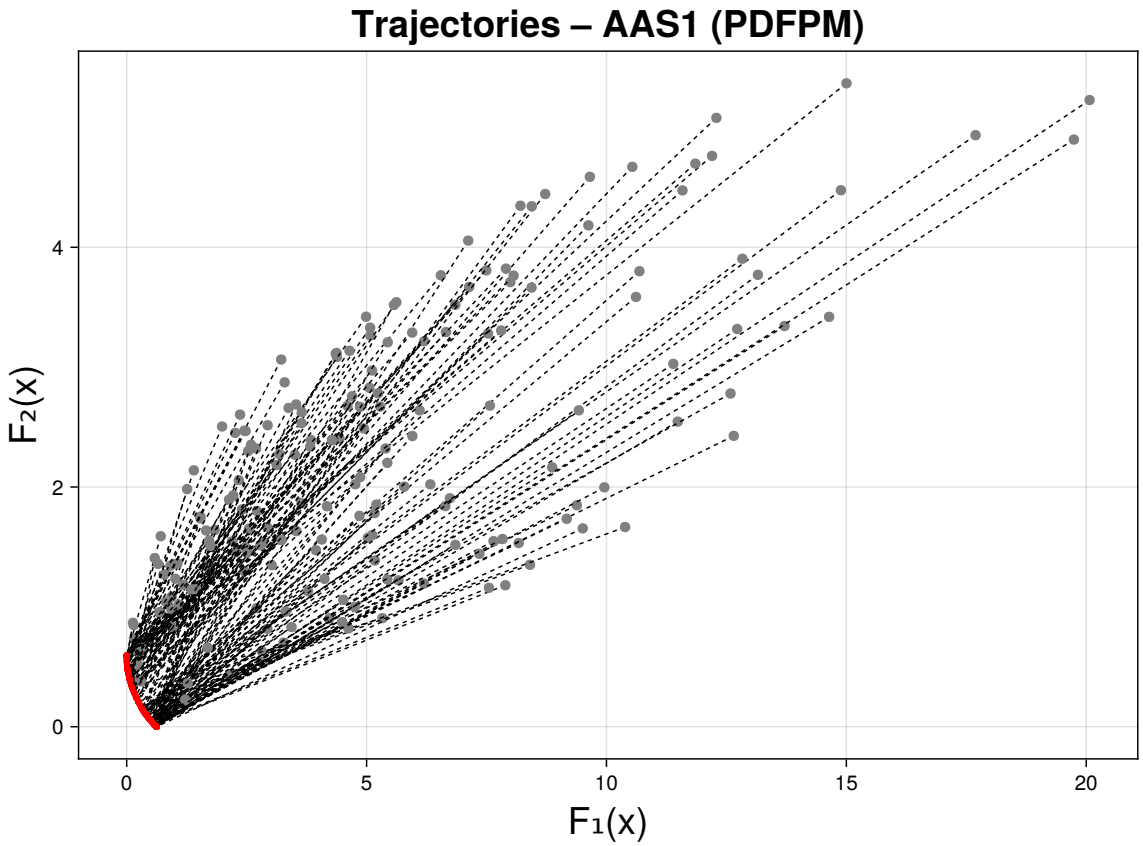
→ **CondG:** the trajectories and Pareto front approximation.

Pareto Fronts: AAS1



→ **ProxGrad:** the trajectories and Pareto front approximation.

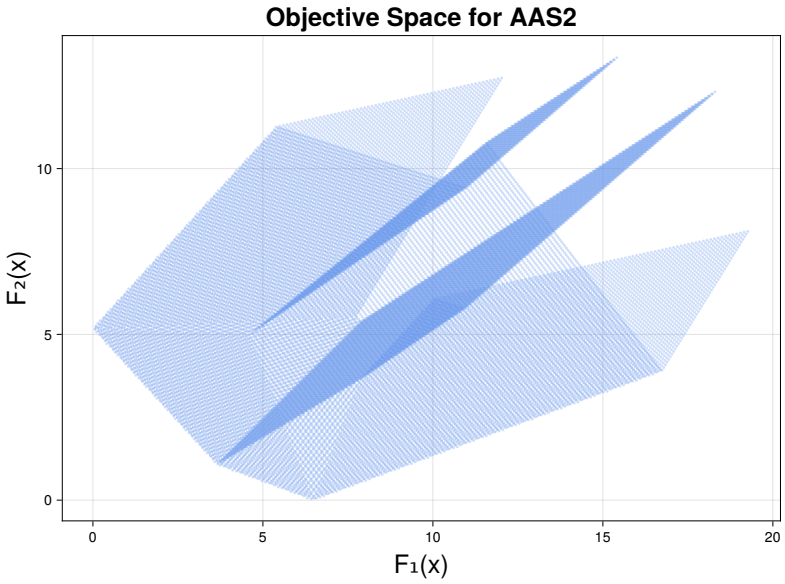
Pareto Fronts: AAS1

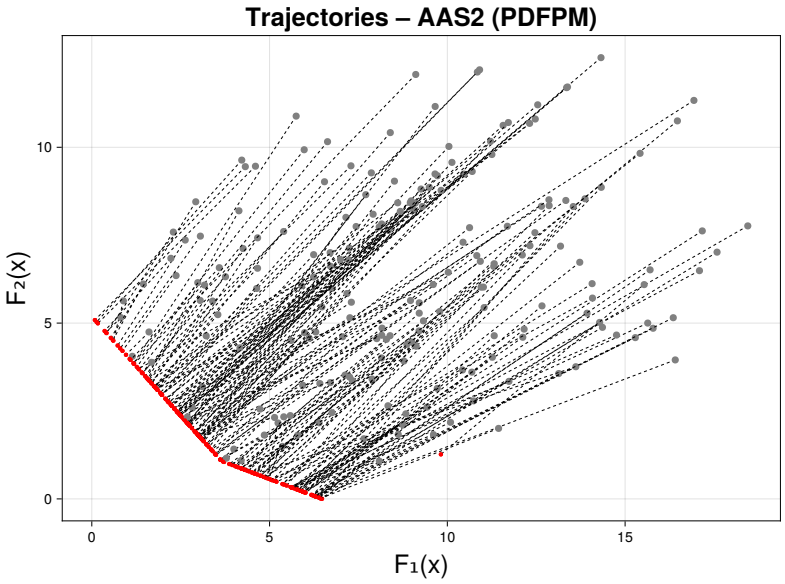


→ **PDFPM:** the trajectories and Pareto front approximation.

We can observe that our method obtained a more significant part of the Pareto frontier.

Pareto Fronts: AAS2 (Fully Hölder Case)



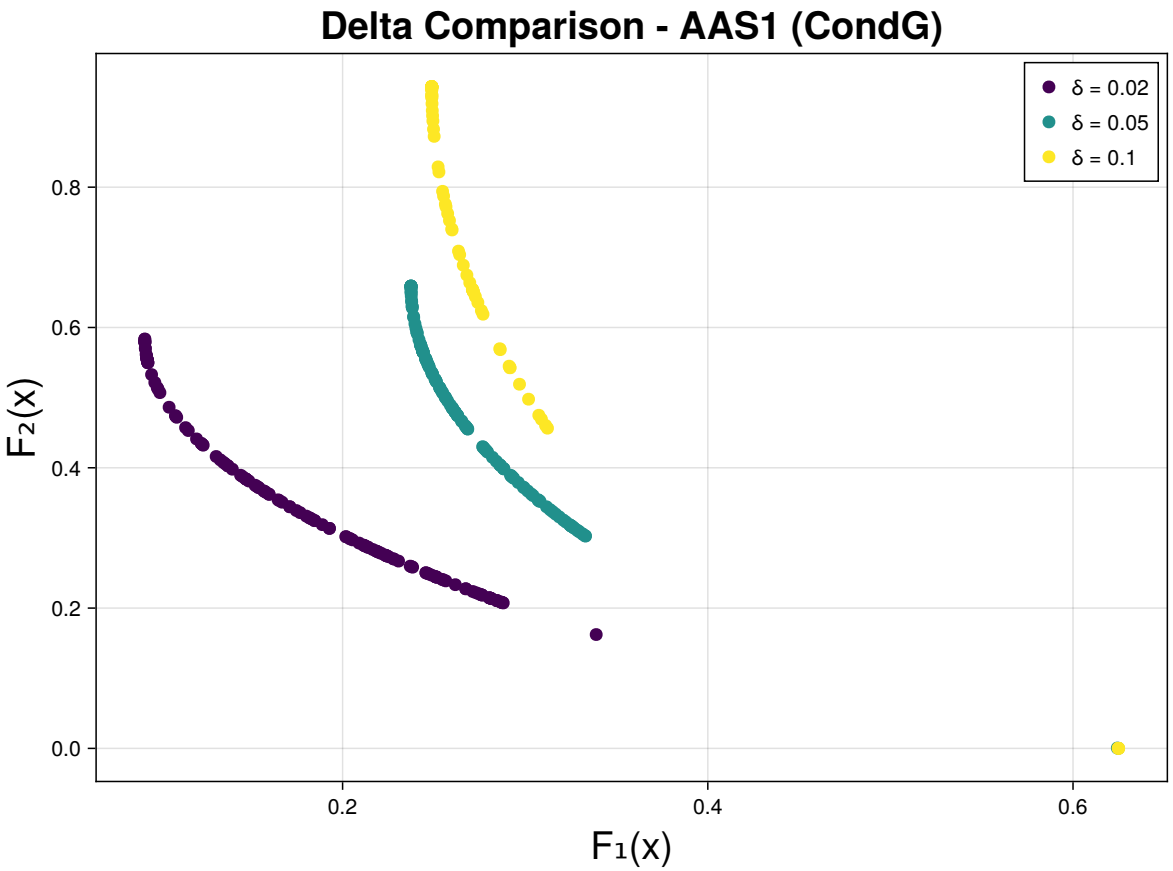


Remark:

- → CondG and ProxGrad did not converge on AAS2 in our experiments.
- → In this problem, both objectives have **Hölder gradients**.
- → PDFPM remains applicable and successfully approximates the Pareto front.

Effect of Uncertainty

The following figures illustrate the effect of increasing uncertainty levels δ on the reconstructed Pareto fronts.



→ **AAS1 – CondG:** reconstruction for increasing values of δ.

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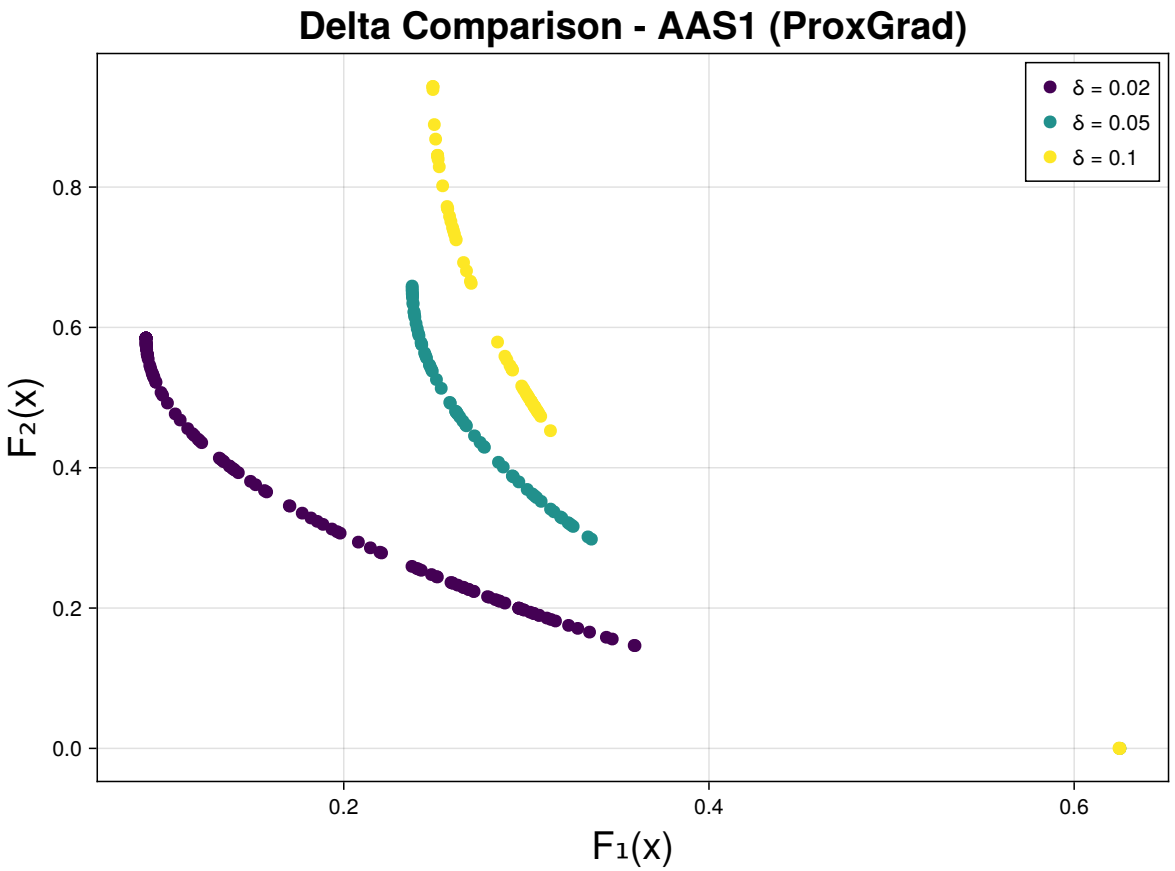

$\rightarrow$ **AAS1 – ProxGrad:** reconstruction for increasing values of δ.

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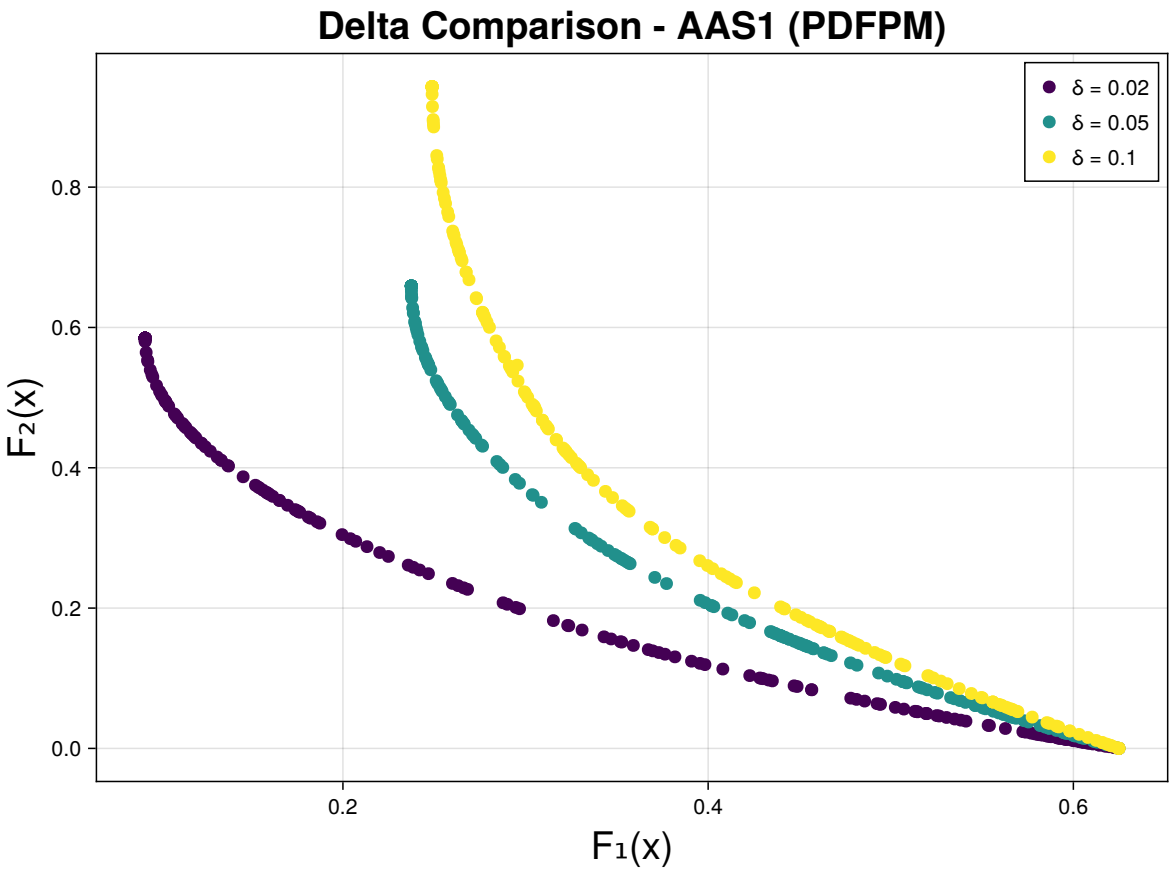

→ **AAS1 – PDFPM:** reconstruction for increasing values of δ.

Effect of Uncertainty

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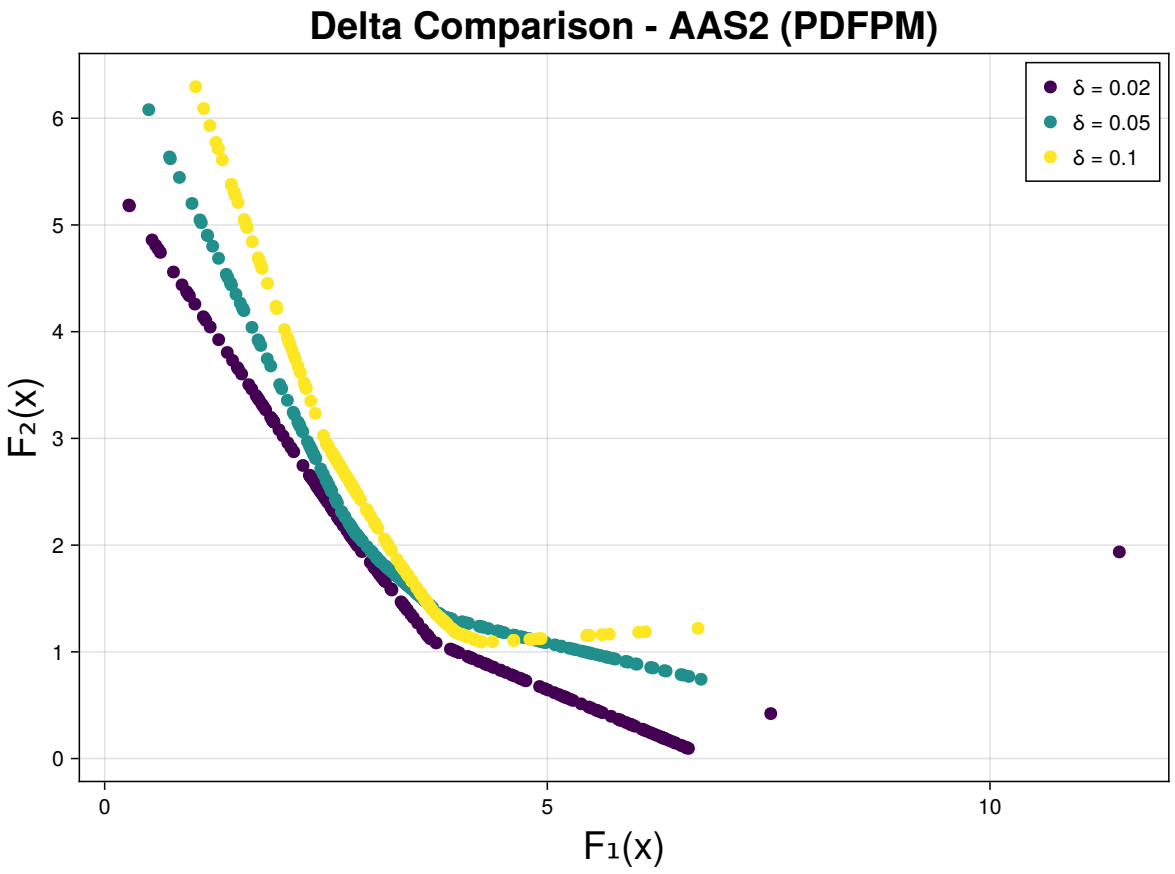


→ **AAS2 – PDFPM only:** CondG and ProxGrad did not converge.

Performance Profiles: Experimental Setup

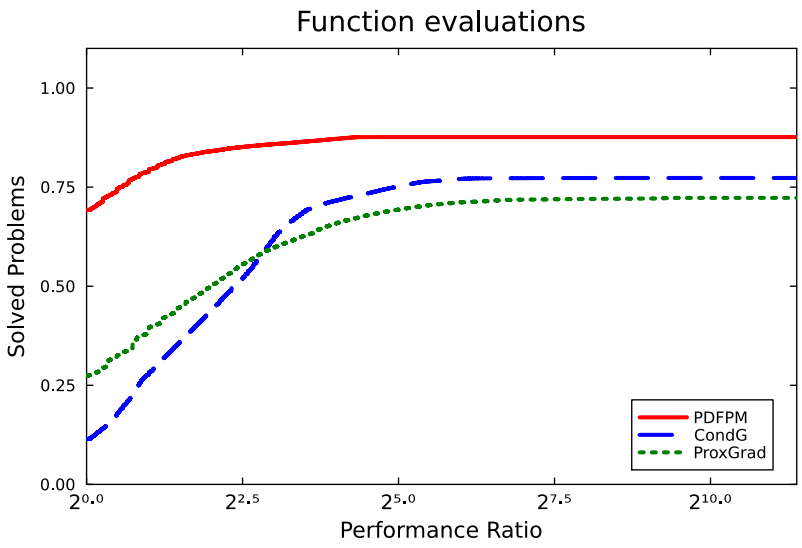
To evaluate the robustness and efficiency of the proposed method, we compare **PDFPM** with two first-order methods: **ProxGrad** and **CondG**.

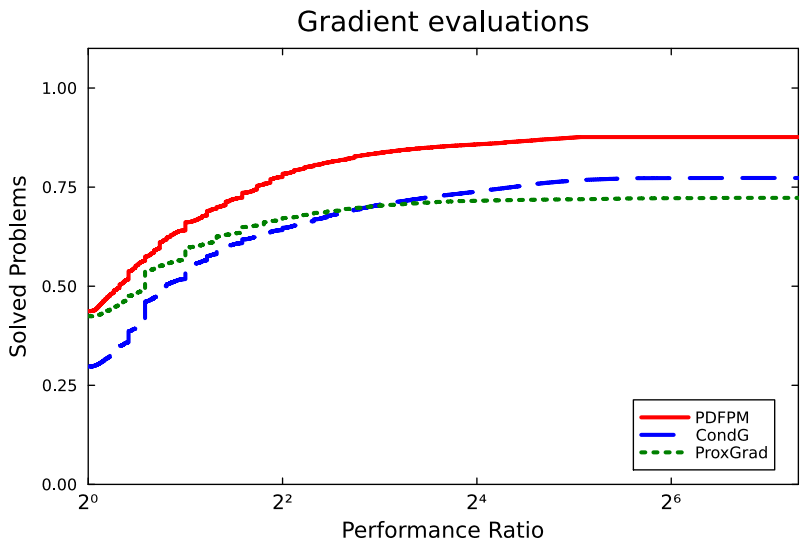
- → **Test set:** multiobjective problems commonly used in the literature.
- → **Problems:** 70 problems with different dimensions and objectives.
- → **Uncertainty levels:** $\delta \in \{0.0,\ 0.02,\ 0.05,\ 0.1\}$.
- → **Runs:** 200 random starting points per instance.
- → **Total instances:** 56,000 runs (problems $\times$ uncertainty levels $\times$ starting points).

Performance is assessed using **performance profiles** based on:

- → number of function evaluations,
- → number of gradient evaluations,
- → iteration count,
- → total execution time.

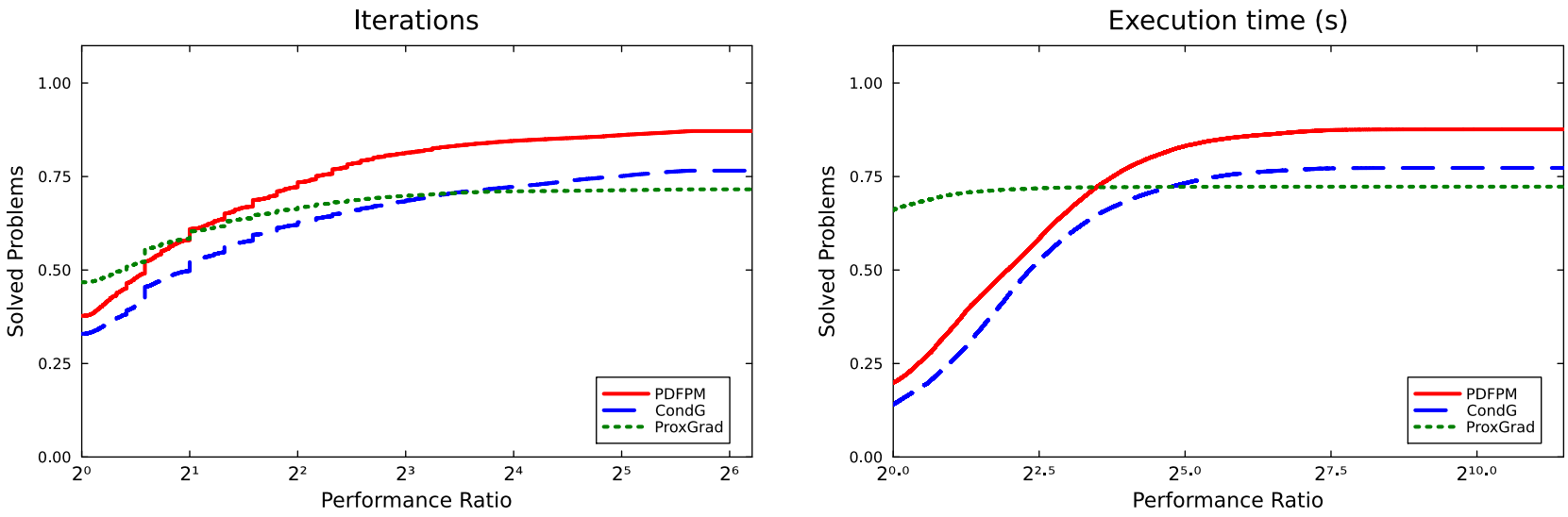
Performance Profiles: Results





$\rightarrow$ PDFPM is more efficient and more robust in terms of **function and gradient evaluations**.

Performance Profiles: Results



→ ProxGrad is often faster in terms of iterations and time, but **PDFPM solves the largest fraction of instances overall**.

Conclusions

→ PDFPM works well for multiobjective problems where:

- all f_j have Lipschitz gradients;
- all f_j have Hölder gradients;
- or a mix of Lipschitz and Hölder gradients.

→ Complexity results are consistent with the literature under Hölder conditions.

- Coinciding with the complexity of the method proposed by Calderón et al.[8] when $p = 1$.
- For $m = 1$, recovers classical complexity of quadratic regularization methods proposed by Martínez[9].

[8] Calderón, L., Diniz-Ehrhardt, M.A., Martínez, J.M.: On high-order model regularization for multiobjective optimization. Optimization Methods and Software 37(1), 175–191 (2022)

[9] Martínez, J.M.: On high-order model regularization for constrained optimiza- tion. SIAM Journal on Optimization 27(4), 2447–2458 (2011)

Conclusions

→ The method does not need previous information on the Lipschitz and Hölder constants.

→ Numerical results:

- Excellent success rates in robust multiobjective problems for all levels of uncertainty tested.
- Stable performance even with Hölder gradients.
- Competitive with other methods for composite multiobjective optimization. (ProxGrad and CondG)

Thank you very much for your attention!

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QR code to access the Paper (arXiv)

QR code to access the Repository with the codes (GitHub)